

Gilberto G. Briscoe-Martinez

PHD CANDIDATE, COMPUTER SCIENCE · NASA SPACE TECHNOLOGY GRADUATE RESEARCHER

Boulder, CO, USA

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Education

University of Colorado Boulder

Boulder, CO, USA

PH.D. IN COMPUTER SCIENCE

Aug. 2021 – May 2027 (expected)

- Advisor: Prof. Alessandro Roncone, HIRO Group.
- Research: fail-active manipulation, embodiment-aware policies, and contact-rich control.
- Funded by the NASA Space Technology Graduate Research Opportunity (NSTGRO) fellowship.

The University of Texas at Austin

Austin, TX, USA

B.S. IN AEROSPACE ENGINEERING

Aug. 2017 – May 2021

- Undergraduate research in the **Human Centered Robotics Lab (HCRL)**, Aerospace Engineering & Engineering Mechanics, advised by Prof. Luis Sentis (Jan. 2018 – May 2021).
- Contributor to the UT Austin Villa@Home RoboCup team (AI-HRI 2019).

Publications

Underline denotes equal contribution. Authors are listed in the order they appear on the publication.

CONFERENCE PAPERS

1. Gilberto G. Briscoe-Martinez, Yaashia Gautam, Rahul Shetty, Anuj Pasricha, Marco M. Nicotra, and Alessandro Roncone, “Moving On, Even When You’re Broken: Fail-Active Trajectory Generation via Diffusion Policies Conditioned on Embodiment and Task,” in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, 2026.
2. Yaashia Gautam, Gilberto Briscoe-Martinez, Adhitya Mohan, Nataliya Nechyporenko, Alessandro Roncone, and Marco M. Nicotra, “Compliant Explicit Reference Governor for Contact Friendly Robotic Manipulators,” in *IFAC World Congress*, 2026.
3. Gilberto Briscoe-Martinez, Anuj Pasricha, Ava Abderezaei, Santosh Chaganti, Sarath Chandra Vajrала, Sri Kanth Popuri, and Alessandro Roncone, “Exploring How Non-Prehensile Manipulation Expands Capability in Robots Experiencing Multi-Joint Failure,” in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, Oct. 2024, pp. 2943–2950. [\[doi\]](#)
4. Rishi Shah, Yuqian Jiang, Haresh Karnan, Gilberto Briscoe-Martinez, Dominick Mulder, Ryan Gupta, Rachel Schlossman, Marika Murphy, Justin Hart, Luis Sentis, and Peter Stone, “Solving Service Robot Tasks: UT Austin Villa@Home 2019 Team Report,” in *AAAI Fall Symposium on Artificial Intelligence and Human-Robot Interaction (AI-HRI)*, Nov. 2019. [\[pdf\]](#)

PREPRINTS

1. Ava Abderezaei, Nataliya Nechyporenko, Joseph Miceli, Gilberto Briscoe-Martinez, and Alessandro Roncone, “Physically Accurate Rigid-Body Dynamics in Particle-Based Simulation,” *arXiv preprint arXiv:2603.14634*, Apr. 2026. [\[arXiv\]](#) [\[code\]](#)

PATENTS

1. Pawankumar Jajoo, Elizabeth R. Goodfriend, and Gilberto G. Briscoe-Martinez, “Computer-Implemented System and Method for Administering an Examination,” U.S. Patent **US 10,339,827** (Issued). [\[link\]](#)

Awards

Jul. 2022 – **NASA Space Technology Graduate Research Opportunity (NSTGRO) Fellowship**, NASA —

USA

Present Grant 80NSSC22K1211

2021 **Graduate Research Fellowship**, Department of Computer Science, University of Colorado

Boulder, CO, USA

Boulder

Mar. 2016 **National Hispanic Scholar**, National Hispanic Scholar Program

USA

Service

MENTORSHIP

Students mentored:

- Rahul Shetty
- Santosh Chaganti
- Sarath Chandra Vajrala
- Sri Kanth Popuri
- Adhitya Mohan
- Liam Olzac
- Maneesh Gundluru
- Umair Yousaf
- Gyanig Kumar
- Stephen Gillet
- Abraham Saleh
- Klara Nitsche
- Guilherme Valgas Schmidt
- Weston Brach

Research Experience

NASA — National Aeronautics and Space Administration

Boulder, CO, USA (Remote / NASA collaboration)

NASA SPACE TECHNOLOGY GRADUATE RESEARCHER (NSTGRO FELLOW)

Jul. 2022 – Present

- Awarded the NSTGRO fellowship (Grant 80NSSC22K1211) for research on fail-active manipulation in space-relevant scenarios where human rescue is infeasible.
- Collaborate with NASA JSC Robotics on long-term reliability and graceful degradation of robotic systems.

HIRO Group, University of Colorado Boulder

Boulder, CO, USA

GRADUATE RESEARCH ASSISTANT

Aug. 2021 – Present

- Advised by Prof. Alessandro Roncone.
- Lead author on the lab's *fail-active manipulation* research thread: an embodiment-conditioned diffusion policy (DEFT, ICRA'26) that recovers from arbitrary joint failures on a 7-DoF Franka Panda — 99.5% / 46.4% success on unconstrained / constrained motion across 4.7M trajectories & 4.7k failure conditions, with 10/10 real-world task completion.
- Earlier work: non-prehensile failure recovery (IROS'24) showing up to **79%** workspace expansion and **88.9%** task success without a functional end-effector.
- Co-author on CERG (IFAC'26), a collaboration with the ROCC Lab (Prof. Marco M. Nicotra) on compliant contact-rich control.
- Co-author on PBD-R (arXiv:2603.14634), a particle-based simulator with physically accurate rigid-body dynamics for sim-to-real transfer.

Human Centered Robotics Lab (HCRL), UT Austin Aerospace Engineering & Engineering Mechanics

Austin, TX, USA

UNDERGRADUATE RESEARCHER

Jan. 2018 – May 2021

- Multi-year undergraduate research role (Prof. Luis Sentis) spanning perception, planning, and task execution for service robots in human environments.
- Member of the **UT Austin Villa @ Home** RoboCup team — co-authored the AAAI AI-HRI 2019 team report.

Professional Experience

Plus One Robotics

Boulder, CO, USA

ROBOTICS AI/ML RESEARCHER (SUMMER INTERN)

Jun. 2025 – Aug. 2025

- Research on intelligent package sorting for high-throughput warehouse automation.
- Developed vision-driven recovery: detect failed picks, formulate recovery plans, and re-engage without operator intervention.
- Coordinated multi-robot and conveyor systems to keep supply-chain pipelines moving under uncertainty.

Johns Hopkins University Applied Physics Laboratory (JHU APL)

Laurel, MD, USA

ROBOTICS ENGINEER (SUMMER INTERN)

May 2021 – Aug. 2021

- Robotics engineering on APL programs combining hardware integration, controls, and applied research.

METECS

Houston, TX, USA

ROBOTICS SOFTWARE ENGINEER (SUMMER INTERN)

May 2020 – Aug. 2020

- Robotics software for space-station / aerospace applications: multibody dynamics, controls, simulation, and HW/SW integration.